

PHYS 143 – Mechanics and Thermal Physics
Week 6: Gravitation – Supplementary Notes

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These notes hold the derivations and deeper analysis that are not in the lecture. Nothing here is needed to start the tutorial; the starred results are used (without proof) in the lecture.

1 Kepler's second law from angular momentum

The force on a planet of mass m is along the line to the Sun, so its torque about the Sun is $\vec{r} \times \vec{F} = 0$ and the angular momentum $\vec{L} = m \vec{r} \times \vec{v}$ is constant (Lecture 8). In polar coordinates about the Sun the speed perpendicular to $\vec{r}$ is $r\dot{\theta}$, so $L = mr^2\dot{\theta}$. In a short time dt the line sweeps a thin triangle of area

$$dA = \frac{1}{2} r (r d\theta) \Rightarrow \frac{dA}{dt} = \frac{1}{2} r^2 \dot{\theta} = \frac{L}{2m} = \text{constant}.$$

Equal areas are swept in equal times: Kepler's second law. At perihelion and aphelion $\vec{v} \perp \vec{r}$, so $L = mv_p r_p = mv_a r_a$, giving $v_p r_p = v_a r_a$ (*). The law holds for *any* central force, not only $1/r^2$.

2 The shell theorem (outline)

Consider a thin spherical shell of mass M_s and radius a , and a point mass m at distance r from the centre. Slice the shell into rings at polar angle θ measured from the line to m . The potential energy of m with one ring is $-Gm dM/s$, where s is the distance from m to the ring and $dM = M_s \frac{1}{2} \sin \theta d\theta$. With $s^2 = r^2 + a^2 - 2ra \cos \theta$, so that $s ds = ra \sin \theta d\theta$,

$$U = -\frac{GmM_s}{2} \int \frac{\sin \theta d\theta}{s} = -\frac{GmM_s}{2ra} \int_{|r-a|}^{r+a} ds = -\frac{GmM_s}{2ra} [(r+a) - |r-a|].$$

Outside ($r > a$): the bracket is $2a$, so $U = -GmM_s/r$: the shell acts as a point mass at its centre.

Inside ($r < a$): the bracket is $2r$, so $U = -GmM_s/a$, which does not depend on r , and $F = -dU/dr = 0$: no force. (*) For a sphere built from shells, at radius r only the mass within r pulls, as a point at the centre.

g inside the Earth. Uniform density ρ : $M(r) = \frac{4}{3}\pi\rho r^3 = M r^3/R^3$, so $g(r) = GM(r)/r^2 = GMr/R^3 = g_s r/R$. For the real Earth $\rho(r)$ increases toward the core, $g(r) = \frac{G}{r^2} \int_0^r 4\pi r'^2 \rho(r') dr'$, and g stays near 9.8 m/s^2 down to the core boundary.

3 The potential energy $U = -GMm/r$

The force on m by M is $\vec{F} = -(GMm/r^2)\hat{r}$. The work done by gravity moving m from r_1 to r_2 along the radial line is

$$W = \int_{r_1}^{r_2} \left(-\frac{GMm}{r^2}\right) dr = GMm \left(\frac{1}{r_2} - \frac{1}{r_1}\right) = -\Delta U \Rightarrow U(r) = -\frac{GMm}{r} + \text{const},$$

and the constant is chosen to make $U \rightarrow 0$ as $r \rightarrow \infty$. (The work depends only on the end points, so the force is conservative, Lecture 5.) **Near the surface:** for $r = R + h$,

$$U(R+h) - U(R) = GMm \frac{h}{R(R+h)} = \frac{GMm}{R^2} h \left(1 + \frac{h}{R}\right)^{-1} \approx mg_s h \left(1 - \frac{h}{R} + \dots\right),$$

so $mg_s h$ is the first term; the relative error is about h/R (6% at 400 km, matching 3.93 against 3.70 MJ/kg).

4 Escape speed

Mechanical energy is conserved: $\frac{1}{2}mv^2 - GMm/r = E$. Escaping to $r \rightarrow \infty$ requires $K \geq 0$ there. The minimum case has $K \rightarrow 0$ and $U \rightarrow 0$, so $E = 0$ and at the surface $\frac{1}{2}mv_{\text{esc}}^2 = GMm/R$:

$$v_{\text{esc}} = \sqrt{\frac{2GM}{R}} = \sqrt{2g_s R} = \sqrt{2} v_{\text{circ}}.$$

For the Earth $\sqrt{2(9.82)(6.371 \times 10^6)} = 11.2 \text{ km/s}$. The result is independent of the direction of launch (the energy argument ignores the path) provided the path does not hit the planet, and independent of m . Because U depends only on r , the same argument gives the escape speed from any r : $\sqrt{2GM/r}$.

5 Energy of a circular orbit

Circular orbit: $GMm/r^2 = mv^2/r$, so $K = \frac{1}{2}mv^2 = GMm/2r$. Then

$$E = K + U = \frac{GMm}{2r} - \frac{GMm}{r} = -\frac{GMm}{2r} = -K = \frac{1}{2}U.$$

A satellite that loses energy $\Delta E < 0$ has r decrease, $K = -E$ increase: drag speeds it up. The virial-type relation $2K = -U$ holds for any $1/r^2$ circular orbit.

6 The ellipse: geometry and energy

Geometry. An ellipse is the set of points whose distances to the two foci add to the constant $2a$. With foci at $\pm c$ on the major axis, $b^2 = a^2 - c^2$, $e = c/a$, and $b = a\sqrt{1 - e^2}$. In polar coordinates about a focus, $r(\vartheta) = \frac{a(1 - e^2)}{1 + e \cos \vartheta}$, giving $r_p = a(1 - e)$ at $\vartheta = 0$ and $r_a = a(1 + e)$ at $\vartheta = \pi$ (so $r_p + r_a = 2a$).

Energy and a . At perihelion and aphelion $\vec{v} \perp \vec{r}$. Conservation of E and of L between the two:

$$\frac{1}{2}v_p^2 - \frac{GM}{r_p} = \frac{1}{2}v_a^2 - \frac{GM}{r_a}, \quad v_p r_p = v_a r_a.$$

Substituting $v_a = v_p r_p / r_a$ and simplifying gives $v_p^2 = 2GM/r_a (r_p / (r_p + r_a))$, and then

$$E = \frac{1}{2}v_p^2 - \frac{GM}{r_p} = -\frac{GM}{r_p + r_a} = -\frac{GM}{2a} \quad (\text{per unit mass}), \quad \text{so} \quad E = -\frac{GMm}{2a}.$$

Vis-viva. With $E = \frac{1}{2}mv^2 - GMm/r = -GMm/2a$ at any point of the orbit:

$$v^2 = GM \left(\frac{2}{r} - \frac{1}{a} \right).$$

Checks: circle ($r = a$): $v^2 = GM/r$; perihelion $v_p^2 = \frac{GM}{a} \frac{1+e}{1-e}$; aphelion $v_a^2 = \frac{GM}{a} \frac{1-e}{1+e}$; $r \rightarrow \infty$ with $E = 0$ ($a \rightarrow \infty$): $v = \sqrt{2GM/r}$.

7 Kepler's third law for ellipses

The area of an ellipse is $A = \pi ab$. From Section 1, $dA/dt = L/2m$, so the period is $T = A/(dA/dt) = 2m\pi ab/L$. At perihelion $L = mv_p r_p$ with $v_p^2 = \frac{GM}{a} \frac{1+e}{1-e}$ and $r_p = a(1-e)$:

$$L^2 = m^2 \frac{GM}{a} \frac{1+e}{1-e} a^2 (1-e)^2 = m^2 GM a (1-e^2) = m^2 GM \frac{b^2}{a}.$$

Hence $T^2 = \frac{4\pi^2 m^2 a^2 b^2}{L^2} = \frac{4\pi^2 a^2 b^2 a}{GM b^2} = \frac{4\pi^2}{GM} a^3$. The semi-major axis a takes the place of r in the circular formula.

8 Weight on the rotating Earth at latitude ϕ

A body at latitude ϕ moves in a circle of radius $R_E \cos \phi$ with acceleration $\omega^2 R_E \cos \phi$ toward the axis. Resolving along the local vertical, the apparent weight is

$$n \approx F_g - m\omega^2 R_E \cos^2 \phi,$$

so the reduction is largest at the equator (0.35%, from $\omega^2 R_E = 0.0339 \text{ m/s}^2$) and zero at the poles. The scale also reads a slightly small value at the equator because the Earth is flattened by its rotation (the equatorial radius is larger), another $\sim 0.2\%$ effect.

9 Limits of Newtonian gravitation

• **Extended bodies:** the point-mass law applies to spheres of spherically symmetric density; otherwise integrate over the mass distribution (tides arise from the difference of the Moon's pull across the Earth). • **Two-body motion:** both bodies orbit their common centre of mass; for $m \ll M$ it is at the larger body, so Kepler's third law with M holds. For two comparable masses, $T^2 = 4\pi^2 a^3 / G(M+m)$. • **Strong gravity:** setting $v_{\text{esc}} = c$ gives the (heuristic) radius $R_S = 2GM/c^2$, $\approx 3 \text{ km}$ for the Sun; the correct treatment needs general relativity (Young §13.8).

Exercises

1. Use $v_p r_p = v_a r_a$ and energy conservation to show that $E = -GMm/2a$ for $r_p = a(1-e)$, $r_a = a(1+e)$ (do the algebra in Section 6 yourself).
2. Show that $v_{\text{esc}} = \sqrt{2} v_{\text{circ}}$ at every r .
3. Using $r(\vartheta)$, check that the area $\frac{1}{2} \int r^2 d\vartheta$ for $e = 0$ is πa^2 .
4. For two masses m_1, m_2 orbiting their centre of mass at separation d , show $T^2 = 4\pi^2 d^3 / G(m_1 + m_2)$.